

"Files are collections of data made by users". The file system is very important for users because it helps store and organize their data. Files have some key properties:

Q1

1. **Long-term existence** – Files stay on the disk even after the user logs off.
2. **Sharable between processes** – Files have names and can be shared with other programs by controlling access permissions.
3. **Structure** – Files can be arranged in folders to show how they are connected to each other."

This makes it easier for users to manage and use files.

File systems keep important information about each file.

Common file operations are:

Q2

- **Create:** Make a new file
- **Delete:** Remove a file
- **Open:** Access a file
- **Close:** Finish working with a file
- **Read:** Get data from a file
- **Write:** Add or change data in a file"

"File structure includes four important terms:

1. **Field** – A single piece of information.
2. **Record** – A group of related fields.
3. **File** – A collection of records.
4. **Database** – A collection of files.

STRUCTURE TERMS

Field

A single piece of information

- It's a basic element of data.
- Contains a single value. (Name)
- Fields can have fixed size or variable length size.

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Record

A record is a group of fields that are related to each other.

Q3

- It is treated as one unit by programs.
- Like fields, records can be **fixed** or **variable** in size.
- Example: A record could be all the fields about one person: Name, Age, Address

File

A file is a collection of similar records.

- It's treated as a single entity (one unit).
- Files can be identified by a name.
- Programs can have restrictions on who can open or change files (access control).
- Example: A file may contain many records, like a list of people and their information.

Database

A database is a collection of related files.

- It clearly shows how the data is connected (relationships).
- Databases are designed to be used by different programs and users.
- Example: A library database might have files for books, authors, and borrowers.

File Management System

A file management system helps users manage their data. Its main tasks are:

- **Meeting user needs** for storing and organizing data.
- **Making sure data is correct** and valid.
- **Improving performance** so file operations are fast.
- **Supporting different storage devices**, like hard drives or USBs.
- **Reducing the risk of losing data** by protecting it.
- **Providing a standard way** for programs to read and write data.
- **Allowing multiple users** to use files at the same time in multi-user systems."



Minimal User Requirements **Q5**

- **Creating, deleting, reading, writing, and editing files.**
- **Having controlled access** to other users' files if needed.
- **Deciding what kind of access** other users can have on their files.
- **Moving data** from one file to another.
- **Back up and recovering files** in case of problems.
- **Finding files by name**, not by numbers or IDs.



Device Drivers

Q6

- Device drivers are the **lowest level** of the operating system.
- They **communicate** directly with hardware devices.(printer,keyboard etc.)
- Responsible for **start and complete I/O operations**.
- Responsible for completing I/O requests.
- Considered part of the operating system.

Basic File System

Q7

- Also called as **physical I/O level**.
- It communicates with external devices like disks and tapes.
- It manages **blocks of data**, their placement on storage, and buffering in memory.
- It does not understand the data content or file structure.
- It's considered part of the operating system.



Basic I/O Supervisor

- "The basic I/O supervisor controls all input/output (I/O) processes.
- It starts and ends file I/O operations.
- It manages control structures for device I/O, scheduling, and file status.
- It chooses the correct device for I/O operations.
- It schedules disk and tape usage to improve performance.
- It assigns I/O buffers and allocates secondary memory.
- This is part of the operating system."

Logical I/O

- **Allowing access to records** in a file.
- Provides general-purpose record I/O .
- **Keeping basic information about files**, such as their size or type."



Access Method

- Access method is the closest part of the file system to users.
- Provides a standard interface for applications.
- Different methods depend on file structure and data processing.

File Organization and Access

"**File organization** is how records are structured for access.

Key criteria for choosing file organization:

- Fast access
- Easy updates
- Storage efficiency
- Simple maintenance
- Reliability

The priority depends on the application."



File Organization Types

There are 5 common types of file organization:

1. The pile
2. The sequential file
3. The indexed file
4. The indexed sequential file
5. The direct (hashed) file

Q9

1- The Pile

"The **pile** is the simplest file organization."

- Data is stored in the **order it arrives**.
- Each record is a separate set of data.
- The **main goal is to collect and save data**.
- Searching requires checking all records.

2- The Sequential file

"The **sequential file** is the most common file type.

- Records have a fixed format.
- A key field uniquely identifies each record.
- It is often used in batch processing.
- Easy to store on both tape and disk."

3- Indexed Sequential File **Q10**

"An **indexed sequential file** uses an **index** to allow random access.

- It includes an **overflow file** for extra records.
- It reduces the **time to find a record**.
- Multiple index levels improve access efficiency."

4- Indexed Files

"An **indexed file** allows access only through indexes.

- Records can have **variable lengths**.
- **Exhaustive index**: One entry for each record.
- **Partial index**: Entries only for selected records.
- Used in systems where **timely information is important**, like **airline reservations** or **inventory control**."

5- Direct or Hashed File

"A **direct or hashed file** allows direct access to any record using its address.

- Uses **hashing** on the key value.
- Ideal for situations needing **fast access** and **fixed-length records**.
- Examples: **Directories, pricing tables, schedules, name lists.**"

B- Trees

"**B-Trees** are balanced trees where all branches have the same length.

- They are a standard way to organize **indexes** in databases.
- Commonly used in **OS file systems**.
- They allow **efficient search, insert, and delete operations.**"

branches
treedeki
dallanmaları
ifade eder.

B-Tree is defined by its minimum degree **d** and follows these rules:

- Each node can have up to **$2d - 1$ keys** and **$2d$ children**.
- Nodes (except root) have at least **$d - 1$ keys** and **d children**.
- The root has at least **1 key** and **2 children**.
- All leaves are on the same level.
- A non-leaf node with **k pointers** has **$k - 1$ keys**.